rag -- 2026-11-01

5 new paper(s) · topic: Retrieval-augmented generation: retrieval quality, grounding and faithfulness, evaluation benchmarks, and RAG system design.

Worth reading in full

[P1] Factual Yet One-Sided: Stance Poisoning and Evidence Balancing in Retrieval-Augmented Generation -- P1 introduces a practical, low-cost middleware that directly tackles stance poisoning--a critical, under-explored failure mode for RAG--while providing detailed quantitative reductions and latency analysis, making it the most broadly relevant read for researchers interested in retrieval quality, grounding, and system design.

Themes

Several papers focus on improving the grounding of retrieved evidence before generation. P1 inserts a lightweight Stance-Balancing Middleware that classifies retrieved documents with a RoBERTa-large-MNLI classifier and enforces a fixed pro-con-neutral quota, cutting stance deviation by up to 0.350 points without extra LLM calls [P1]. Likewise, P3 augments prompts with retrieved passages from domain-specific knowledge bases (Wikipedia, slang lexicon, feminist literature) and shows that feminist literature yields the strongest explanatory benefit, raising recall on Italian implicit-misogyny detection from 0.838 to 0.865 and dramatically improving human-preferred explanations [P3]. Both approaches treat retrieval as a controllable pre-processing step, but P1 targets bias mitigation while P3 targets coverage of implicit background knowledge.

Evidence selection mechanisms that respect input budgets are explored in P4 and P5. HyperDoc builds a hypergraph over fine-grained multimodal elements and solves a coverage-maximization problem with a cost-aware greedy algorithm, achieving state-of-the-art results on three long-document QA benchmarks [P4]. RouteDRAG learns a routing policy that predicts when to invoke an Agent and how to control it, reducing Agent calls by ~15 % while improving accuracy by up to 1.28 pp, demonstrating a learned, dynamic alternative to the static selection used in HyperDoc [P5].

Domain-aware structuring of retrieval indexes is examined in P2, which extracts material-Tc pairs from arXiv PDFs and stores them as JSONB metadata for hybrid retrieval. Although hybrid retrieval does not raise overall source recall beyond BM25 (Recall@5 = 47 % vs 41 %), it improves conditional answer accuracy when gold evidence is already retrieved, highlighting a trade-off between structured metadata filtering and raw dense similarity that differs from the purely graph-based or policy-based methods of P4 and P5.

Across the set, evaluation gaps appear: P2 reports only limited benchmark numbers and no end-to-end QA scores; P3 relies on recall and human preference without standard factuality metrics; P4 and P5 focus on accuracy on long-document QA but do not assess stance or bias. This suggests a need for unified benchmarks that jointly measure retrieval quality, grounding fidelity, and bias mitigation in RAG systems.

4 theme sentence(s) checked against the paper summaries.

At a glance

[P1] Factual Yet One-Sided: Stance Poisoning and Evidence Balancing in Retrieval-Augmented Generation · relevance 0.37
The paper introduces a low-cost Stance-Balancing Middleware that detects document stance and enforces a fixed pro/con/neutral quota, substantially reducing one-sided stance shifts caused by factually correct but biased retrievals in RAG systems.

[P2] A Domain-Aware Pipeline for Structured Extraction and Retrieval-Augmented Question Answering over arXiv Superconductivity Literature · relevance 0.36

The paper introduces a domain-aware pipeline that extracts superconducting materials and critical temperatures from arXiv PDFs, stores them as structured metadata, and uses hybrid retrieval-augmented generation, achieving modest gains over regex baselines but limited by noisy extraction and a small gold set.

[P3] RAGing Against Misogyny: Improving Implicit Misogyny Explanations · relevance 0.36
A Retrieval-Augmented Generation (RAG) system improves recall for detecting implicit misogyny in Italian comments and yields human-preferred explanations, especially when using feminist literature as the knowledge source, though gains are limited for English data and several practical constraints remain.

[P4] Hypergraph-Enhanced Evidence Selection for Long-Context Multimodal Document QA · relevance 0.34
HyperDoc introduces a hypergraph-based, budget-aware evidence selection framework that outperforms existing multimodal RAG baselines on long-context document QA tasks.

[P5] RouteDRAG: Learning to Route and Control Agentic Execution for Multimodal Document RAG · relevance 0.34
RouteDRAG learns to predict when to invoke an Agent and how to control its operations, achieving higher accuracy with fewer Agent calls on multimodal document QA tasks.

[P1] Factual Yet One-Sided: Stance Poisoning and Evidence Balancing in Retrieval-Augmented Generation

Puyuan Yang, Shaoxuan Dong, Liujie Wu, Xiaochun Cheng et al. · arXiv 2610.34294v1 · https://arxiv.org/abs/2610.34294v1 · relevance 0.37

TL;DR: The paper introduces a low-cost Stance-Balancing Middleware that detects document stance and enforces a fixed pro/con/neutral quota, substantially reducing one-sided stance shifts caused by factually correct but biased retrievals in RAG systems.

Problem
Retrieval-augmented generation (RAG) can inherit the stance distribution of its retrieved evidence, enabling "stance poisoning" where injected documents are factually correct but push the system toward a one-sided viewpoint--a gap not addressed by prior work focused on factual errors [1].

Method
The authors insert a Stance-Balancing Middleware between the retriever and the generator without altering either component [1]. It first classifies each candidate document as pro, con, or neutral using a locally run RoBERTa-large-MNLI-based classifier (macro-F1 0.889, target-stance accuracy 0.905) [3]. Then it enforces a fixed quota of 2 pro + 2 con + 1 neutral documents and adds a debiasing prompt before passing the evidence to the generator, all without extra LLM calls [1,3,4].

Key results
- The mean topic-level poisoning share in the injected ProCon corpus was 18.9 % and all four tested generators showed stance shifts aligned with the poisoning direction [1].
- Without defense, stance scores rose to +0.230 (± 0.124 - 0.335) for pro-poisoning and -0.197 (± -0.301 - -0.086) for con-poisoning at an 11.9 % poisoning share [4].
- The complete middleware reduced stance deviation by 0.106 - 0.350 points across eight concentration-direction settings; six settings with 11.9 %-18.9 % poisoning remained significant after Holm adjustment (adjusted p <= 0.009) [1,7].
- In the concentration sweep, paired effect-size estimates showed reductions of +0.106 to +0.350 (pro) and +0.111 to +0.313 (con) with Holm-adjusted p <= 0.009 for the higher concentrations [7].
- Cross-model experiments (four generators) moved all eight model-direction combos toward neutral, with seven of eight showing a clear reduction in absolute distance from zero [1].
- Residual pro-side bias in the cross-model setting was small, with point estimates of +0.045 to +0.063 [1,6].

- Stance detection latency for 20 candidate documents was ~1.5 % of median GPU generation time and ~16 % of CPU generation time [1,5].
- The middleware lowered the average number of poisoned documents among the five supplied per question from 1.743 (pro) / 1.424 (con) without defense to 1.181 / 0.986 with balancing, still better than the MMR baseline (1.048 / 0.900) [5].

Limitations
The approach relies on predefined stance categories and accurate labels; missing neutral evidence or misclassifying rhetorical tone can impair selection [6,8]. It has been evaluated only on English data, a single retriever, one question source, and LLM-generated poisoned documents, limiting generality [6]. Small residual pro-side bias remains, and semantic-outlier or graph-based defenses may miss collectively one-sided but factually correct documents [2]. Future work must broaden language, retriever, and content coverage and improve target-aware stance detection.

Sources

[1] arXiv 2610.34294v1.pdf, arXiv abstract

[2] arXiv 2610.34294v1.pdf, page 4

[3] arXiv 2610.34294v1.pdf, page 8

[4] arXiv 2610.34294v1.pdf, page 12

[5] arXiv 2610.34294v1.pdf, page 17

[6] arXiv 2610.34294v1.pdf, page 21

[7] arXiv 2610.34294v1.pdf, page 25

[8] arXiv 2610.34294v1.pdf, page 29

[P2] A Domain-Aware Pipeline for Structured Extraction and Retrieval-Augmented Question Answering over arXiv Superconductivity Literature

Ivan Iorsh, Aleksandr Kapitonov, Ivan Filianin, Matvey Lesonen et al. · *arXiv 2610.34391v1* · *https://arxiv.org/abs/2610.34391v1* · *relevance 0.36*

TL;DR: The paper introduces a domain-aware pipeline that extracts superconducting materials and critical temperatures from arXiv PDFs, stores them as structured metadata, and uses hybrid retrieval-augmented generation, achieving modest gains over regex baselines but limited by noisy extraction and a small gold set.

Problem
Scientific literature in materials physics is expanding faster than researchers can manually inspect it, and existing retrieval tools rely on keyword matching or generic question answering that ignore domain-specific structure in PDFs [1]. Effective downstream tasks therefore require accurate extraction of materials and superconducting transition temperatures (Tc) from the documents [6].

Method
The system builds a thematic superconductivity corpus from arXiv metadata, converts each PDF to text, and applies domain-aware extractors to recover material names, Tc values, article type, dimensionality, unconventional-superconductivity cues, and Debye frequency [3]. Extracted attributes are normalized and stored in PostgreSQL as JSONB payloads linked to the source document [4]. Publications with successful material extraction are chunked into ~1600-character fragments, embedded with a multilingual E5-family model, and indexed for hybrid retrieval that combines dense similarity with structured metadata filtering and domain-aware reranking; a local language model then generates answers grounded in the retrieved context [3][4].

Key results
- Extraction was evaluated on a manually verified gold set of 59 superconducting materials (46 with confirmed experimental Tc, 33 forming a primary positive subset) [5][1].
- On the primary positive subset, domain-aware material-Tc pair accuracy reached 45.5%, compared with 9.1% for a regex baseline [1][7].
- On a 100-question benchmark varying only the retrieval mechanism, BM25 achieved the highest source recall (Recall@5 = 47%) [1]; hybrid retrieval with extracted fields did not improve aggregate source recall (39% vs 41%) [1].
- When gold evidence was already retrieved, hybrid retrieval with fields gave the highest conditional answer accuracy, though only on small subsets (exact numbers not provided) [6][7].
- Matching literature-derived temperatures to an independent theoretical reference set found 200 overlapping materials, of which 20 had a median experimental Tc above the theoretical value [1].

Limitations
The approach is specialized to superconductivity literature, relies on a modest manually verified gold set, and its downstream performance is sensitive to PDF processing and extraction quality; noisy extracted metadata limited the benefit of hybrid field retrieval, intra-document evidence ranking remained weak relative to document recall, and small subsets underlie several conditional metrics [7]. Additionally, there is a trade-off between conservative Tc extraction (leaving fields empty) and aggressive extraction (increasing false positives) [7].

Sources

[1] arXiv 2610.34391v1.pdf, arXiv abstract

[3] arXiv 2610.34391v1.pdf, page 3

[4] arXiv 2610.34391v1.pdf, page 5

[5] arXiv 2610.34391v1.pdf, page 6

[6] arXiv 2610.34391v1.pdf, page 8

[7] arXiv 2610.34391v1.pdf, page 9

[P3] RAGing Against Misogyny: Improving Implicit Misogyny Explanations

Ivana Crescenzi, Arianna Muti, Maarten Sap, Tommaso Caselli et al. · arXiv 2610.35094v1 · https://arxiv.org/abs/2610.35094v1 · relevance 0.36

TL;DR: A Retrieval-Augmented Generation (RAG) system improves recall for detecting implicit misogyny in Italian comments and yields human-preferred explanations, especially when using feminist literature as the knowledge source, though gains are limited for English data and several practical constraints remain.

Problem
Implicit misogyny often relies on background knowledge that is not explicit in the text, making detection and explanation difficult for large language models [1].

Method
The authors build a RAG pipeline that retrieves evidence from three knowledge bases--Wikipedia, a slang lexicon, and feminist literature--and appends the retrieved passages to prompts for a multilingual LLM [2][5][6]. Experiments are run on two datasets (IMPLICIT-MIS for Italian and SBIC+ for English) and evaluated with recall (since all instances are misogynistic) [3][4].

Key results
- On IMPLICIT-MIS, adding the news headline (context) raises recall from 0.838 (Italian, text-only) to 0.865 and from 0.771 (English, text-only) to 0.891 [4].
- RAG does not improve detection performance on SBIC+ (recall not reported) [1].
Human evaluation on 200 instances per language shows RAG explanations are preferred over baseline

explanations in 88 vs 18 cases for IMPLICIT-MIS and 86 vs 27 cases for SBIC+, with additional ties and "neither" selections [5].
- Normalized preference scores indicate feminist literature provides the strongest explanatory benefit (0.408 for IMPLICIT-MIS and 0.538 for SBIC+), outperforming Wikipedia (0.237 and 0.361) and slang (0.255 and 0.500) [5].

Limitations
The study cannot guarantee that all SBIC+ examples are truly implicit, many contain explicit slurs or jokes [7]. Human evaluation is limited to a 100-instance subsample per language, so results are diagnostic rather than exhaustive [7]. Retrieval coverage varies across knowledge sources and languages, affecting comparability of source-level results [7]. Retrieved evidence sometimes lacks pragmatic relevance, being driven by lexical overlap rather than implicit reasoning [7]. Annotations were performed by the paper's authors, which may introduce bias despite anonymization [7]. Experiments rely on a single 8B model; findings may not generalize to other model families or larger scales [7].

Sources

[1] arXiv 2610.35094v1.pdf, arXiv abstract

[2] arXiv 2610.35094v1.pdf, page 3

[3] arXiv 2610.35094v1.pdf, page 4

[4] arXiv 2610.35094v1.pdf, page 5

[5] arXiv 2610.35094v1.pdf, page 7

[6] arXiv 2610.35094v1.pdf, page 8

[7] arXiv 2610.35094v1.pdf, page 9

[P4] Hypergraph-Enhanced Evidence Selection for Long-Context Multimodal Document QA

Wenxin Zhao, Jiachuan Wang, Xinyi Zhu, Jiake Ge et al. · arXiv 2610.35542v2 · https://arxiv.org/abs/2610.35542v2 · relevance 0.34

TL;DR: HyperDoc introduces a hypergraph-based, budget-aware evidence selection framework that outperforms existing multimodal RAG baselines on long-context document QA tasks.

Problem - Long-context multimodal document question answering requires reasoning over dispersed visual and textual evidence while respecting limited input budgets, and existing retrieval-augmented generation pipelines rely on static, fixed-granularity units that cause information loss and suboptimal reasoning performance[1].

Method - HyperDoc builds a document hypergraph that encodes structural, logical, and cross-modal relations among fine-grained elements, then formulates query-aware hyperedge selection as a coverage-maximization problem and solves it with a cost-aware greedy algorithm that adapts to each query's intent and VLM input budget[1][2][4]. The pipeline also includes lightweight evidence-integration modules for reranking and block-level pruning after selection[5].

Key results - Experiments on three representative benchmarks--MMLongBench-Doc (1091 questions, 135 documents, avg. 47.5 pages), Long-DocURL (2325 questions, 396 documents, avg. 85.6 pages), and DocBench (1102 questions, 229 documents, avg. The experiments evaluate HyperDoc's performance on long-context multimodal document QA tasks using three benchmarks with average lengths of 47.5-85.6 pages [6][6].

Limitations - The excerpts do not report any explicit limitations or failure analyses for HyperDoc[7].

Sources

[1] arXiv 2610.35542v2.pdf, arXiv abstract

[2] arXiv 2610.35542v2.pdf, page 2

[4] arXiv 2610.35542v2.pdf, page 6

[5] arXiv 2610.35542v2.pdf, page 7

[6] arXiv 2610.35542v2.pdf, page 8

[7] arXiv 2610.35542v2.pdf, page 12

[P5] RouteDRAG: Learning to Route and Control Agentic Execution for Multimodal Document RAG

Zekai Chen, Hao Zhu, Yu Liu, Xunkai Li et al. · arXiv 2610.36444v1 · https://arxiv.org/abs/2610.36444v1 · relevance 0.34

TL;DR: RouteDRAG learns to predict when to invoke an Agent and how to control its operations, achieving higher accuracy with fewer Agent calls on multimodal document QA tasks.

Problem
Current multimodal document RAG systems often invoke Agentic execution indiscriminately, incurring extra computation and sometimes harming answer quality; the key question is when Agentic execution is truly needed and how to control its subsequent operations [1][2].

Method
RouteDRAG introduces a learning-based pipeline comprising (i) a Unified Document Index that graphs multimodal content, (ii) Adaptive Path Routing that predicts the relative benefit of using an Agent versus a shorter path, (iii) Dynamic Agent Execution that refines tool use based on an evolving workspace, and (iv) Sequential Policy Learning that first trains operation control from observed outcomes and then trains the routing policy using the resulting Agent performance [1][3].

Key results
- Across DocBench and MMLongBench-Doc, RouteDRAG attains 75.86 % and 53.80 % accuracy respectively, surpassing all baselines [4][5].
- It improves over uniform Agent execution by 1.18 pp (DocBench) and 1.28 pp (MMLongBench-Doc) while cutting answering calls by 16.4 % and 14.9 % [4][5].
- The average accuracy gain over shorter-path-only execution is 3.47 pp, with the Agent invoked for only 79.3 % of queries [1].
- RouteDRAG uses 6.24 (DocBench) and 7.02 (MMLongBench-Doc) answering calls per query and processes 46.04 k and 50.99 k tokens per query respectively [6].
- Ablation shows that removing Agent execution drops accuracy to 73.41 %/49.31 %, while uniform Agent execution yields 74.68 %/52.52 %; a low-cost variant (RouteDRAG-Efficient) reaches 74.05 %/50.87 % accuracy with only 11.52 %/16.41 % Agent usage [5][8].

Limitations
The study is limited to single-turn document question answering on two multimodal benchmarks and does not explore multi-turn interactions or broader domain generalization [7].

Sources

[1] arXiv 2610.36444v1.pdf, arXiv abstract

[2] arXiv 2610.36444v1.pdf, page 2

[3] arXiv 2610.36444v1.pdf, page 4

[4] arXiv 2610.36444v1.pdf, page 6

[5] arXiv 2610.36444v1.pdf, page 6
[6] arXiv 2610.36444v1.pdf, page 8
[7] arXiv 2610.36444v1.pdf, page 9
[8] arXiv 2610.36444v1.pdf, page 13